

Computing Basics

The tools of the trade

10/12/22

The age of computing



1964: CDC 6600

“First” supercomputer

Capable of 1 megaFLOP (1 million)

FLOPs= operations (+,-,X) per second



2014: iPhone 6

~1000 megaFLOPS

2022: iPhone 13

~768,000 megaFLOPS

How large are machines now?

■
1 Pf machine
NEU "Discovery"
~30000 cores



iPhone

Area of each signifies
the performance

~200 Pflop machines (2015)

Titan

Sequoia

JUGEEN

MilkyWay-2

Sunway TiahuLight

The current leader (2022)



Frontier Supercomputer at ORNL

More than 600,000 CPU cores and 37,000 GPUs

Fiber optic connections between the cores

Capable of 1,000,000,000,000,000,000 calculations per second

~10,000,000,000,000 times more powerful than the CDC 6600!!!!

Weights approximately 200,000 kg!

Data Storage Yesterday (1956): 5 megabyte IBM data storage drive = 5,000,000 bytes



[viralfhq](#)

A five-megabyte hard drive is shipped by IBM, 1956.

Modern (small) Data Storage



~\$20

128 Gb thumb drive

~25,000 times more storage than the
IBM storage drive

Data Storage Today, 1.5 petabytes data storage system= 1,500,000,000,000,000 bytes

Sony whips up a box to store 1.5 petabytes of data on 10,000 Blu-ray discs

JORDAN NOVET MARCH 10, 2015 2:33 PM

TAGS: BLU-RAY, COLD STORAGE, FACEBOOK, OPEN COMPUTE PROJECT, OPEN COMPUTE SUMMIT, SONY



Above: Sony's new prototype for storing 1.5 PB of data on Blu-ray discs at the Open Compute Summit in San Jose, Calif., on March 10.

This system can store about 300,000,000 times the amount of data that a state-of-the-art IBM (5 megabyte) drive could store in 1956 – see earlier slide

What does this mean?

First teraflop machine appeared in 1996 (ASCI “Red”)

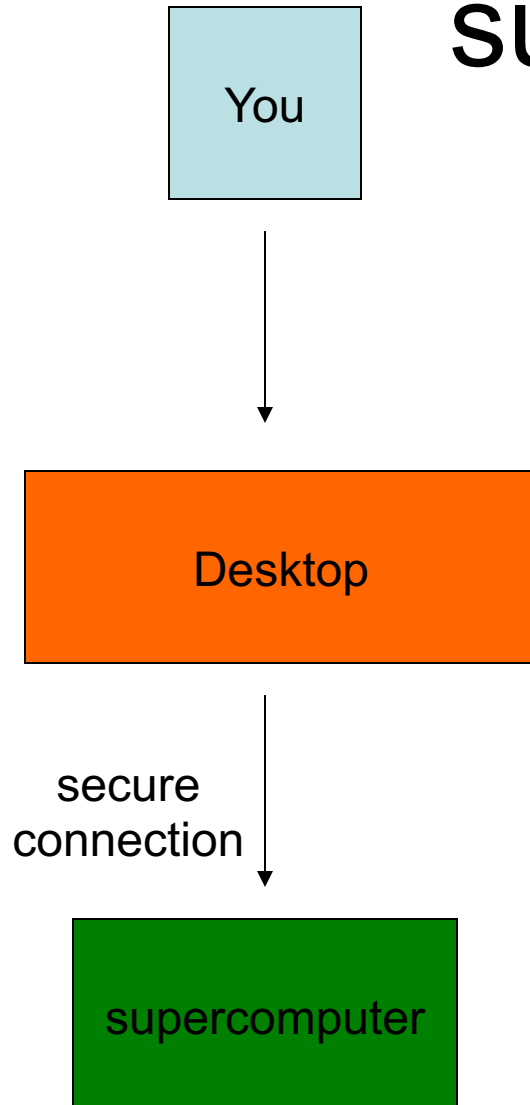
A single GPU can exceed 1 teraflop (~\$2000)

The supercomputing of the early 90s is trivial today

The cost of computation is small enough that anyone can afford to incorporate computational components into their research

**Computers are only tools, we need to ask
the right questions!!!**

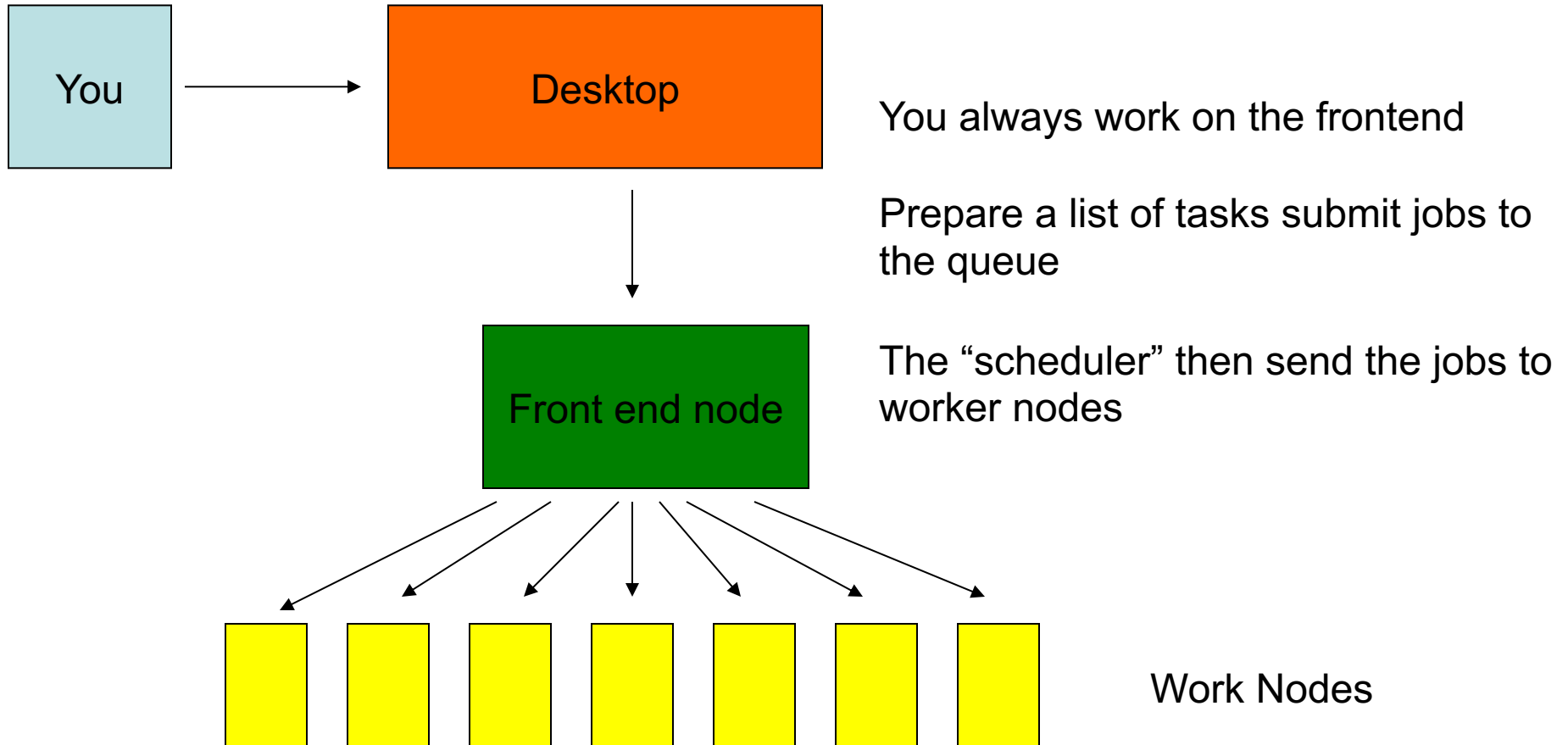
How do you use a supercomputer?



```
pcw2@bgq-fn:~  
Problems and/or questions should be submitted via the problem tracking  
system form: http://helpdesk.rice.edu  
Enter remote host: bgq  
connecting to bgq  
pcw2@bgq's password:   
Last login: Mon Mar 16 14:16:13 2015 from   
-----  
RICE UNIVERSITY BLUEGENE/Q  
=====
```

The screenshot shows a terminal window with a title bar that reads 'pcw2@bgq-fn:~'. The terminal output includes instructions for submitting problems, a prompt for a remote host (bgq), a connection attempt, a password prompt (masked with a box), and a last login timestamp. It then displays a separator line, the text 'RICE UNIVERSITY BLUEGENE/Q', another separator line, and a warning: 'Unauthorized access is prohibited.' followed by 'BGQ,RICE.EDU'. Below this is a link to the helpdesk. Another separator line follows, then a welcome message to Bluegene/Q, a 'Thanks,' message, and the RCSG logo. The prompt '[pcw2@bgq-fn ~]\$' is visible at the bottom.

Where is the work done?



The point?

- Using a terminal is the only real option for using high performance computing resources
- We will use a terminal to work locally on your machine, but the ideas can be used to access larger resources in the future

Using a Terminal

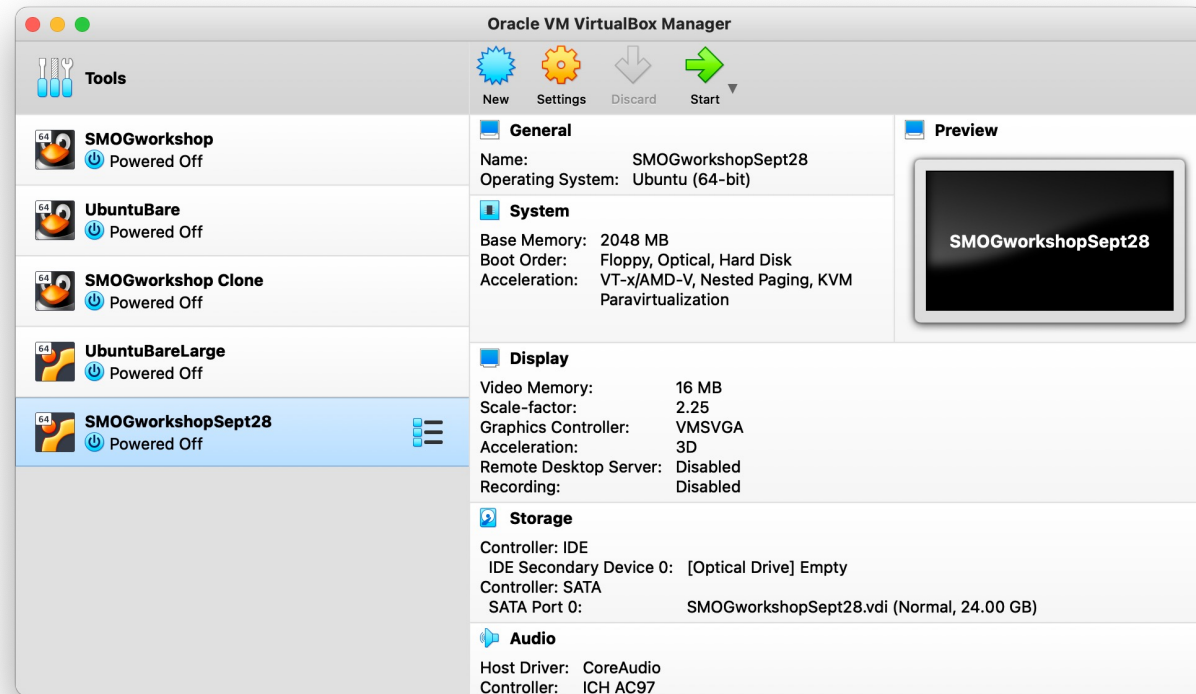
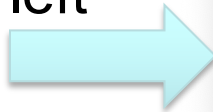
- Today: We will work locally using the terminal
- The terminal is based on **commands**. That is, you can't click on things!
- Here, we are going to use the terminal inside of a virtual machine

Virtual Machines

- A VM is a piece of software that allows you to install a second operating system and use it as if it were an actual computer
- For example:
 - You may have a Windows machine
 - You can install VM software
 - When in the VM, you can install a Linux operating system
 - Then, use the Linux-based VM, even though you are on a Windows machine

Starting the Linux Virtual Machine

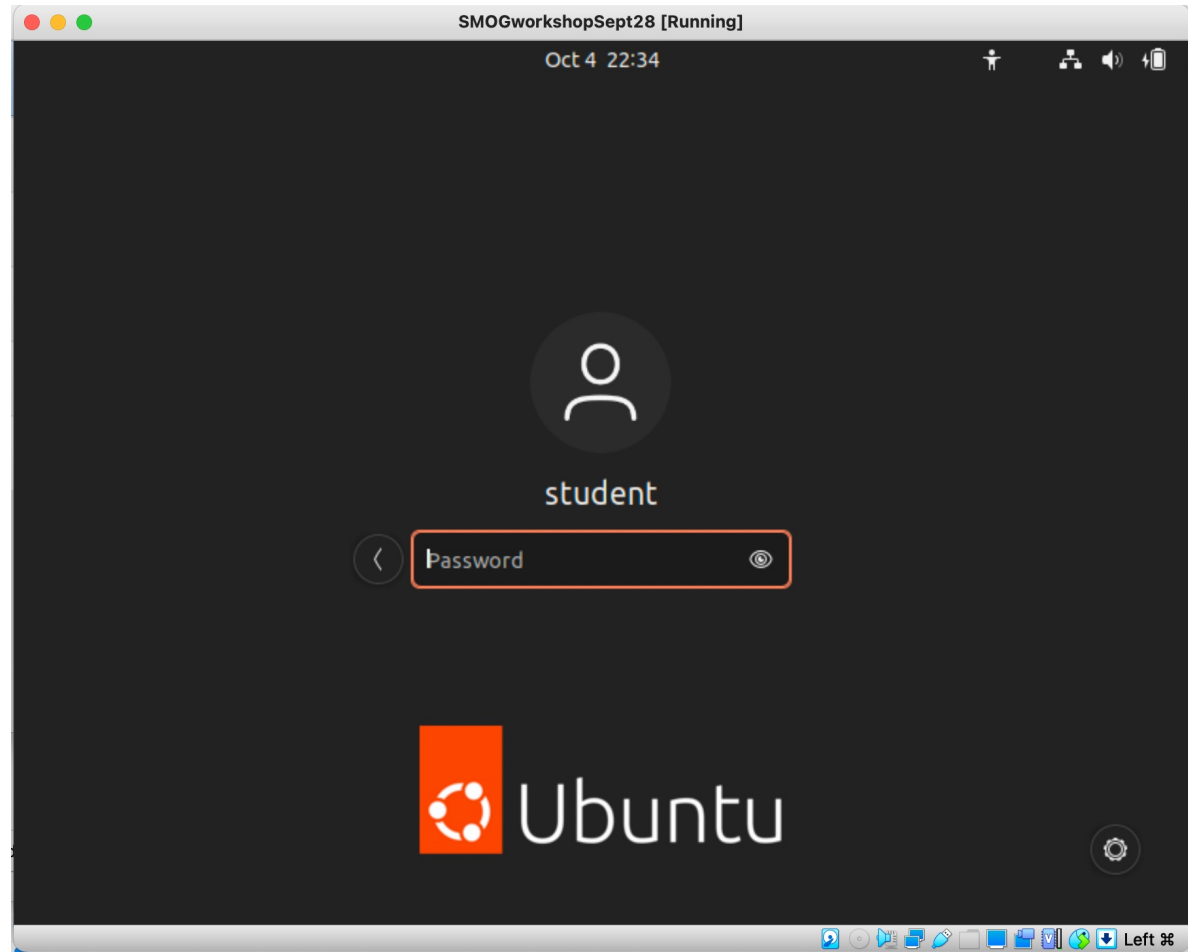
- Start VirtualBox
- If you already imported the image, then simply click on the VM image on the left



If the VM launches, you will see an Ubuntu based OS and a prompt to login

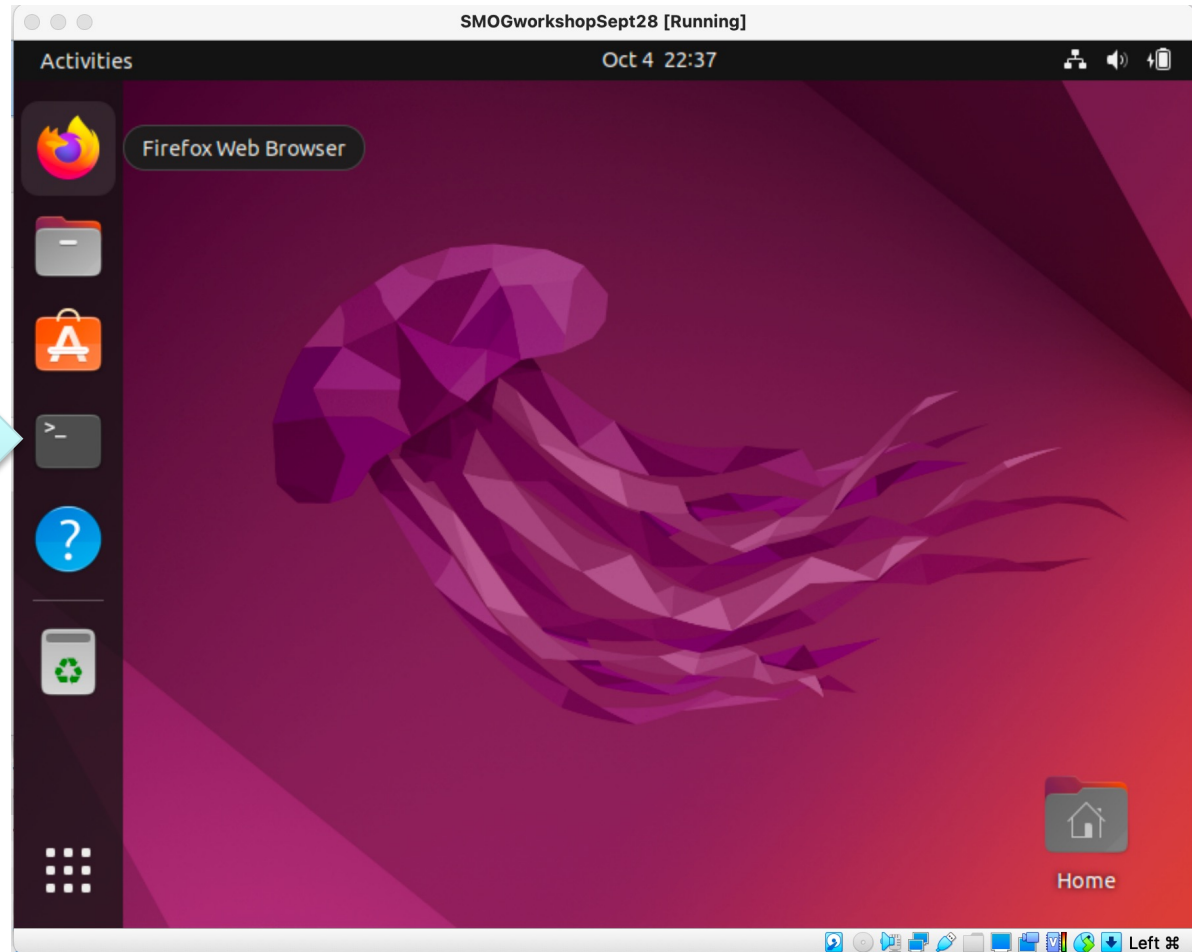
Login with
Username: student
Password: smog

You may be automatically logged in. That is fine. If you get logged out, you can use the credentials above.

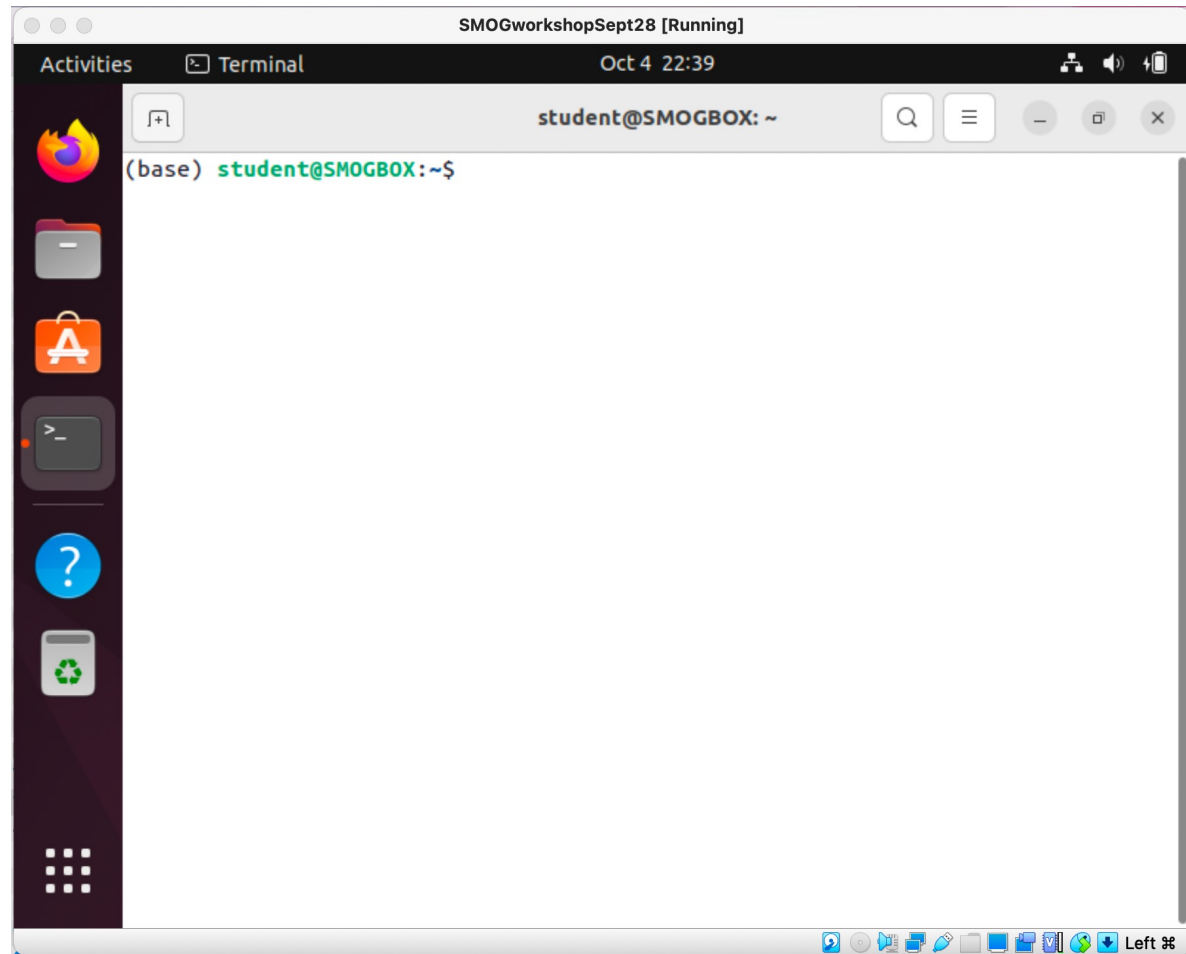


Once logged in, you are in Linux.

Open the terminal by clicking on the gray box



You are now in the terminal



Trying useful Linux commands

- Linux is command based
 - i.e. You enter text (commands) and hit “enter” key if you want something to happen
- Common Commands (follow along)
 - Make a directory: `mkdir TEMPDIR`
 - Change directory (going into TEMPDIR) “`cd TEMPDIR`”
 - List contents of directory: `ls`
 - You should see nothing
 - View a file `Stuff.txt`: “`gedit Stuff.txt &`”
 - Enter text, save and close
 - Remove file `Stuff.txt`: “`rm Stuff.txt`”
 - Get out of directory: “`cd ..`”
 - Remove directory: “`rmdir TEMPDIR`”
 - Show where you are: “`pwd`”

Programs we will use tomorrow

All of these tools are already installed in the VM.

- VMD: Visual molecular dynamics (available in the VM)
 - To start session, type “vmd” in the Terminal
- SMOG 2
 - This will be used to generate models from structure
- OpenSMOG/OpenMM
 - This will be used to perform simulations